

## Some definitions

*set* is finite sets implemented as Lists.

$Subst := set (Var * Exp)$

$Equation := set ((exp * exp) + (sexp * sexp))$

$Problem := set Equation$

$Tuple := Subst * Problem$

$-\langle\_ \rangle : exp \rightarrow Subst \rightarrow exp$

$-\langle\_ \rangle : sexp \rightarrow Subst \rightarrow sexp$

$-\langle\_ \rangle : Problem \rightarrow Subst \rightarrow Problem$

$-\ @ \_ : Subst \rightarrow exp$

$dom : Subst \rightarrow set Var$

$smatch : Tuple \rightarrow Tuple \rightarrow Prop$

*smatch* steps are classified into three

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### Congruence steps

$$\begin{aligned} \text{smatch\_app\_congr} : (s \approx t \approx s' \approx t') \in P \rightarrow \\ \text{smatch}(S, P) \text{ } (S, P + (s \approx s') + (t \approx t') \setminus (s \approx t \approx s' \approx t')) \end{aligned}$$

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### $\sigma$ -min steps

$$\begin{aligned} \text{smatch\_subst\_comp} : (s[\sigma \circ \tau] \approx s'[\rho]) \in P \rightarrow \\ \text{smatch } (S, P) \ (S, P + (s[\sigma][\tau] \approx s'[\rho]) \setminus (s[\sigma \circ \tau] \approx s'[\rho])) \end{aligned}$$

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## Instantiation case

$$\begin{aligned} \text{smatch\_inst} : (s =_? \ X) \in P \rightarrow \\ X \notin \text{lhvars\_of} \ P \rightarrow \\ S' =_{\sigma_{\min}} S \gg \{(X, s)\} \rightarrow \\ \text{smatch} \ (S, P) \ (S', (P \setminus (s =_? \ X)) \langle \{(X, s)\} \rangle) \end{aligned}$$

# Preservation statement

## Validity

$$\text{valid\_tuple } (S, P) := \underline{(\text{dom } S) \cap (\text{vars\_of } P) = \phi}$$

## Matching solution

$$\text{match\_sol } SI \ (S, P) := \frac{(s =_? t) \in P \rightarrow s =_{\sigma_{min}} t \langle SI \rangle \wedge \underline{\exists S', S \gg S' =_{\sigma_{min}} SI}}{}$$

## Preservation theorem

$$\begin{aligned} \text{match\_pres} : & \text{valid\_tuple } T \rightarrow \\ & (\text{dom } SI) \cap (\text{lhvars\_of } P) \rightarrow \\ & \text{smatch } T \ T' \rightarrow \\ & \text{match\_sol } SI \ T' \rightarrow \\ & \text{match\_sol } SI \ T \end{aligned}$$

# Proof proceeds by case analysis on *smatch*

## Inst case

$H_1 : (\text{dom } S) \cap \text{vars\_of } P = \{\}$

$H_2 : (\text{dom } SI) \cap \text{lhvars\_of } P = \{\}$

$H_3 : X \notin \text{lhvars\_of } P$

$H_4 : \text{match\_sol } SI (S \gg \{(X, s)\}, (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle)$

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$\text{match\_sol } SI P$

## Inst case (Unfolded *match\_sol*)

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$$H_4 : \forall s' \ t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s =_{\sigma_{\min}} t \langle SI \rangle$$

$$H_5 : \exists S', SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

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$$\forall s_0, t_0, (s_0 =_? t_0) \in P \rightarrow s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle \quad \wedge$$

$$\exists S', SI =_{\sigma_{\min}} S \gg S'$$



## Inst case (Unfolded *match\_sol*)

$$H_1 : (\text{dom } S) \cap \text{vars\_of } P = \{\}$$

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$$H_3 : X \notin \text{lhvars\_of } P$$

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Second conjunct is solvable associativity property of substitution composition.

Inst case continued..

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$$s_0, t_0 : \mathbf{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

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Case analysis :  $\{s_0 =_? t_0 = s =_? X\} + \{s_0 =_? t_0 \neq s =_? X\}$

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$s =_{\sigma_{min}} SI @ X$  is not immediate

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3.  $(S \gg \{(X, s)\}) @ X = s$ , from  $H_1$  and  $H_6$



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4.  $((S \gg \{(X, s)\}) \gg S') @ X = ((S \gg \{(X, s)\}) @ X) \langle S' \rangle = s \langle S' \rangle$ , from 3 and lookup property

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6.  $s =_{\sigma_{min}} s \langle S' \rangle$ , from 5

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Property mentioned in 5:

$\forall X\ S\ S'\ Sl, X \notin dom\ S \rightarrow X \notin Sl \rightarrow S \gg S' =_{\sigma_{min}} Sl \rightarrow S' @ X =_{\sigma_{min}} X$

Case 2.1:  $s_0 =_? t_0 \neq s =_? X$ , when  $X \in \text{vars\_of } t_0$

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$$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$$

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4.  $Y \in (\text{vars\_of } s) \rightarrow Y \notin \text{dom } SI$ , from  $H_2$  and  $H_6$
5.  $SI @ X =_{\sigma_{\min}} s$ , same as in case 1.
6.  $SI =_{\sigma_{\min}} \{(X, s)\} \gg SI$ , from 4 and 5.

Case 2.1:  $s_0 =_? t_0 \neq s =_? X$ , when  $X \in \text{vars\_of } t_0$

$$H_1 : (\text{dom } S) \cap \text{vars\_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars\_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars\_of } P$$

$$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle SI \rangle$$

$$H_5 : SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

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$$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$$

1. We can show  $\underline{s_0 =_? t_0 \langle \{(X, s)\} \rangle} \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle$ .
2. 1 in  $H_4$  gives us  $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \langle SI \rangle$
3.  $s_0 =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg SI$ , due to substitution composition property.
4.  $Y \in (\text{vars\_of } s) \rightarrow Y \notin \text{dom } SI$ , from  $H_2$  and  $H_6$
5.  $SI @ X =_{\sigma_{\min}} s$ , same as in case 1.
6.  $SI =_{\sigma_{\min}} \{(X, s)\} \gg SI$ , from 4 and 5.
7.  $t_0 \langle SI \rangle =_{\sigma_{\min}} t_0 \langle \{(X, s)\} \rangle \gg SI$

Case 2.2:  $s_0 =_? t_0 \neq s =_? X$ , when  $X \notin \text{vars\_of } t_0$

$$H_1 : (\text{dom } S) \cap \text{vars\_of } P = \{\}$$

$$H_2 : (\text{dom } SI) \cap \text{lhvars\_of } P = \{\}$$

$$H_3 : X \notin \text{lhvars\_of } P$$

$$H_4 : \forall s' t', (s' =_? t') \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{\min}} t' \langle SI \rangle$$

$$H_5 : SI =_{\sigma_{\min}} (S \gg \{(X, s)\}) \gg S'$$

$$s_0, t_0 : \text{exp}$$

$$H_6 : (s_0 =_? t_0) \in P$$

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$$s_0 =_{\sigma_{\min}} t_0 \langle SI \rangle$$

1. Apply  $H_4$ .
2.  $(s_0 =_? t_0) \in (P \setminus (s =_? X)) \langle \{(X, s)\} \rangle$  is provable by knowing  $H_6$ ,  $X \notin \text{vars\_of } t_0$ , and  $s_0 \neq s$  and  $t_0 \neq X$ .